

Topic 7-2

T cells help other cells by telling them how to eliminate pathogens via release of cytokines

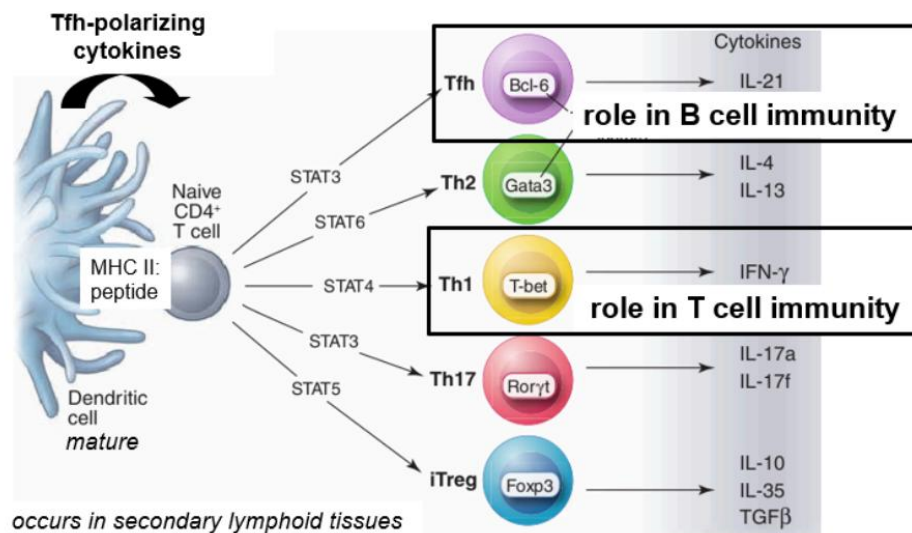
Do not destroy pathogens directly

Many different types of Effector Th cells

For a naive T cell to know which type of helper cell to become - all dictated by the mature dendritic cell (DC)

Mature DC 1) binds to naive T helper cells and 2) secretes polarizing cytokines telling that naive T helper cell to become an effector T helper 1 cell (yellow one)

Different Types of Effector Th cells



Role of Effector T Helper 1 cells is to eliminate intracellular bacteria that are living in the phagosome/phagolysosome of macrophages

(Recall: those intracellular bacteria that are in the cytoplasm of the phagocyte will be destroyed by the effector T cells)

Effector Th1 cells are ineffective here

Why are macrophages predominantly the cell infected by intracellular bacteria?

Bc many intracellular bacteria require iron for growth/survival in host cells

Macrophages are important in normal iron homeostasis - they phagocytose RBC (dn need to know for exam)

Some intracellular bacteria are still destroyed by lysosomal contents - so there is always gonna be some exogenous bacterial peptide being presented on MHC class 2 molecules by the infected macrophage

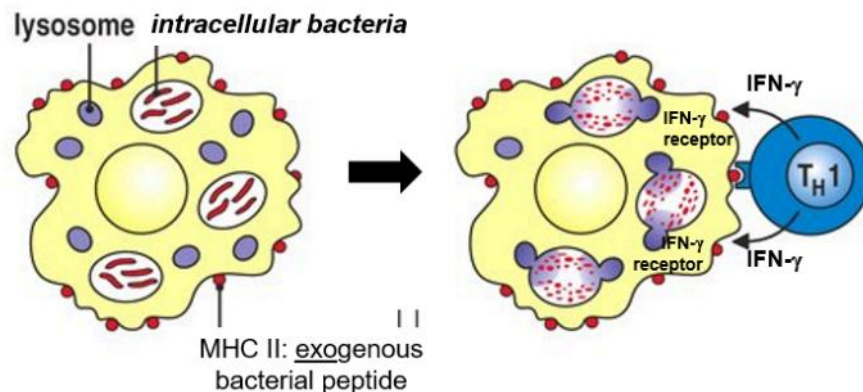
Effector Th1 cells bind to peptide on the MHC class 2 molec, and secrete cytokine: IFN- (gamma)

IFN- γ will bind to interferon gamma receptor present on the surface of the macrophage

this sends activation signal to the infected macrophage to tell it to increase lysosome-phagosome fusion -> more acids and lysosomal enzyme released onto that intracellular bacteria

IFN- γ also tells macrophage to increase production of toxic enzymes (NADPH oxidase and nitric oxide synthetase) to incrition of toxic oxygen metabolites and nitric oxide .

1. Effector Th1 cells secrete IFN- γ that activates infected macrophages → increases lysosome-phagosome fusion



However, that is not destructive for 100% of the intracellular bacteria

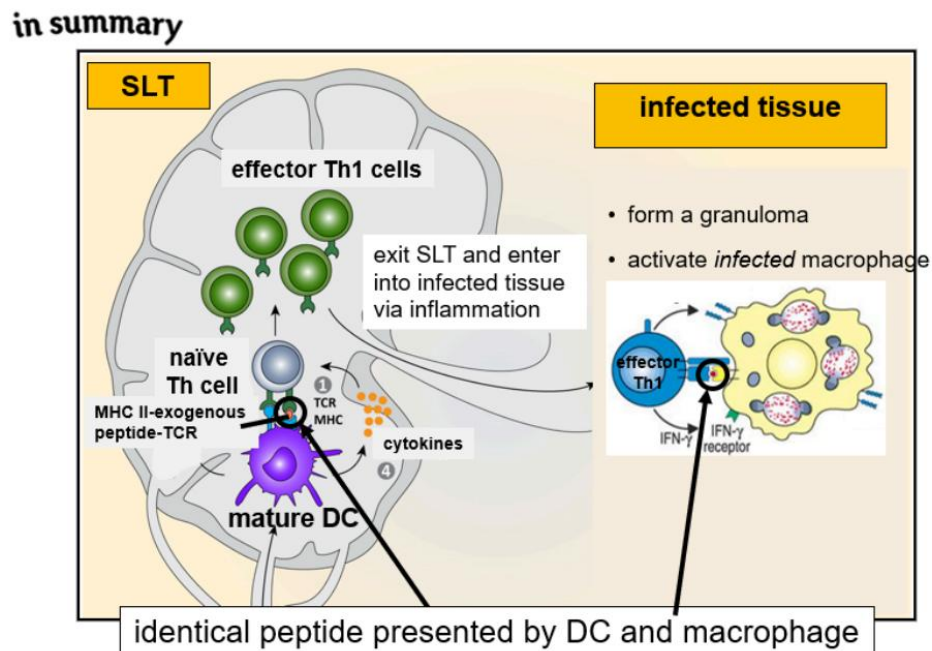
Another role of Effector Th1 cell is to wall off/ surround the infected macrophages in the tissue and suffocate the bacteria

Called granuloma

Blocks entry of nutrients and O₂ -> starves the intracellular bacteria

Kills infected macrophages but doesn't always kill the bacteria (ie tuberculosis -> TB remains latent for life - we cannot 100% kill these pathogens)

In summary:



Mature DC presents peptide that naïve cell binds to - MDC secretes polarizing cytokines onto naïve T helper cell telling it to become effector Th1 cell

Daughter cells become effective Th1 cells enter SLT and enter into infected tissue via inflammation (same as effector CTL)

Once in tissue, it binds to infected macrophage (MHC class 2) and it produces IFN- γ which binds to interferon gamma receptor .

- Sends activation signal to macrophage to produce more toxic O₂ metabolites + nitric oxide , and increase phagosome-lysosome fusion ->

Aldo forms granuloma to hopefully suffocate the infected macrophages (not always effective)

Topic 7-3

Immature dendritic cells reside in blood & tissues

It contains peripheral tissue and secondary lymphoid tissue

Pathogen can enter a dendritic cell by:

Phagocytosis (extracellular bacteria)- peptides presented on MHC class 2 molecs

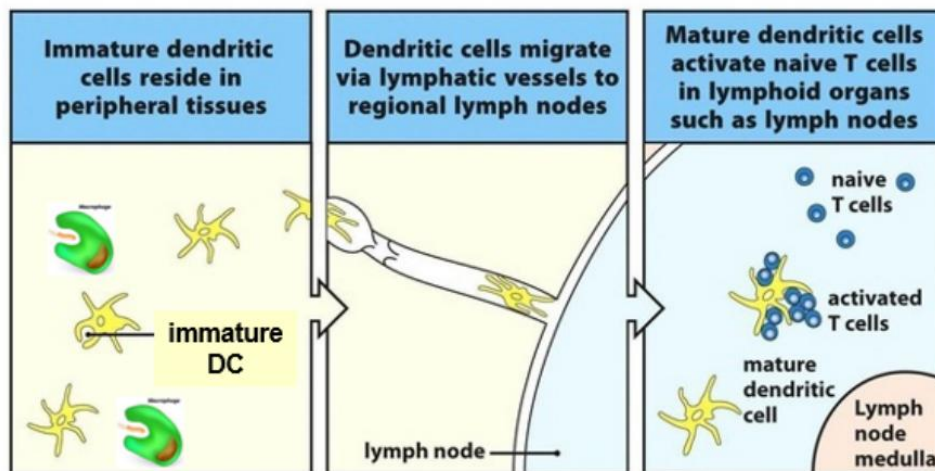
2) Viral entry or intracellular bacteria (DC infected) - peptides presented on MHC class 1 molecs

Immature DC internalize pathogens in peripheral tissues

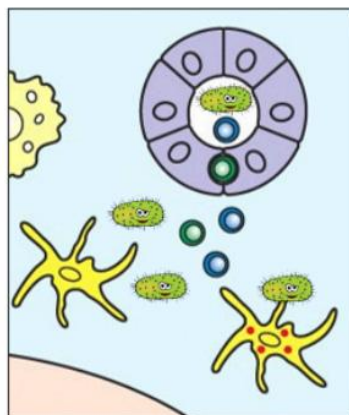
Some DC exit tissues and migrate via lymphatics to lymph node

DCs mature during migration & become APC (Antigen-Presenting Cell)

Mature DC enters T cell area of Lymph Node (LN) and present its peptides on MHC molecules to naive T cells



Some immature DC are resident in SLT (secondary lymphoid tissue) & meet pathogens entering the spleen (via blood) and MALT (via mucosa)



- DC “internalize” incoming pathogens.
- Present peptides on MHC molecules to **naïve T cells** circulating through SLT.

Naive T cell will interact w/ a peripheral DC at the lymph node, or they will interact w/ a resident DC at the spleen, lymph node, or the MALT tissue

1 naive T cell circulates thru a lymph node every 12-24hrs

Macrophages that are present in peripheral tissue do not exit the peripheral tissue

In mucosal areas - say a cut in your mouth

Pathogens enter and encounter immature DC under epithelial cell layer

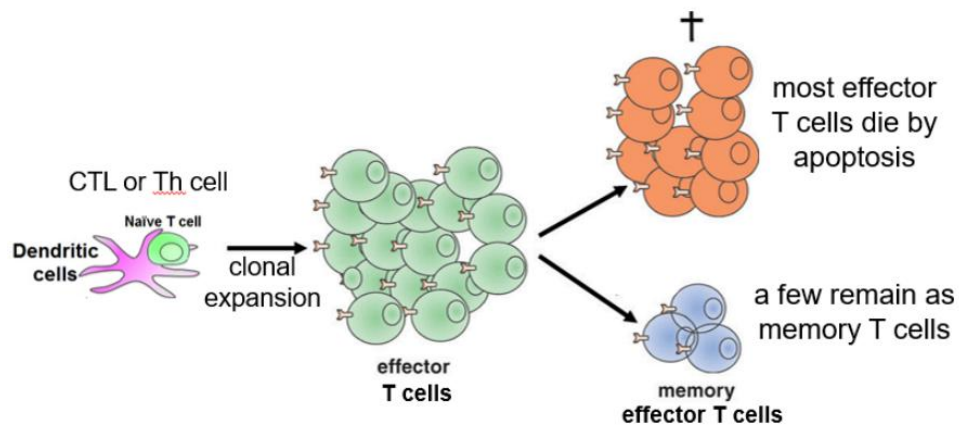
Immature DC take up pathogen and some will enter lymphatics and enter regional lymph node

Some pathogens that entered can also enter MALT tissues via the M cell (ie tonsils in mouth)

Naive T cells will be responding to pathogens in both MALT tissues and lymph nodes

This is bc most pathogens enter our body via our mouth .

T cell immunity always initiated in SLT by mature dendritic cell to naive T cell



- Most effector

T cells will die by

apoptosis

Clearing our pathogen

- Few will remain as

memory effector

Cells

There is no diff btw

The memory T cell

effector T cell

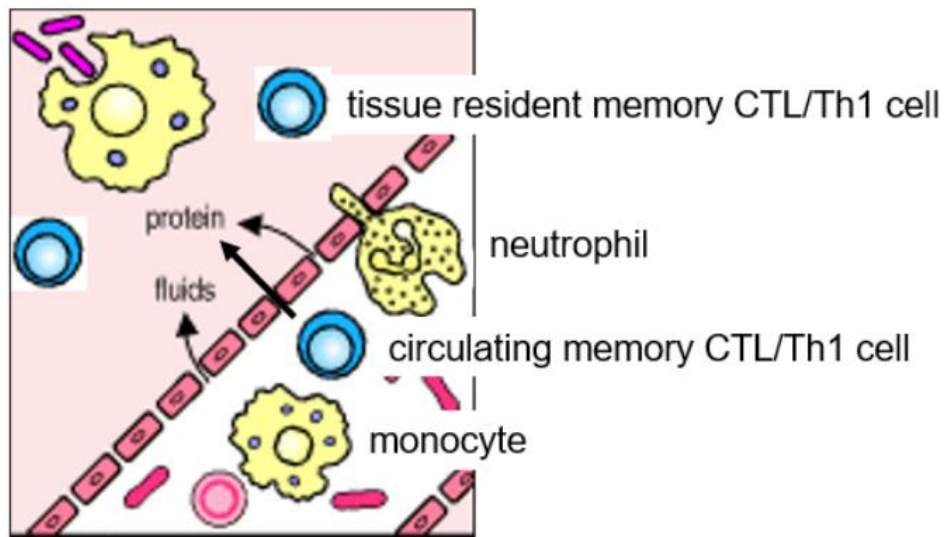
We know how memory T cells are regulated

T cell

Response is more effective (we already generated those effector T cells so they can respond ASAP to re-expose of the identical pathogen)

Reside in SLTs

Upon re-infection, memory T cells can act alongside immunity (neutrophils, monocytes, immature DC)



(next couple slides are review slides)

Topic 8-1

B cell immunity

B CELL IMMUNITY LEARNING OBJECTIVES:

By the end of this section, you should know:

1. The kinetics of antibody production following primary exposure vs. re-exposure to a protein vs. non-protein antigen.
2. What cells are generated in a primary vs. memory immune response?
3. How antibodies function.
4. The process of somatic hypermutation and class switching.
5. What B cell components provide longterm immune protection?
6. Passive immunity in a newborn and an adult.

B cell immunity results in production of antibodies to help eliminate extracellular pathogens and inactive soluble proteins (e.g toxins)

Each B cell expresses multiple copies (approx 100,000) of a B cell receptor (BCR) with specificity to one unique antigenic epitope

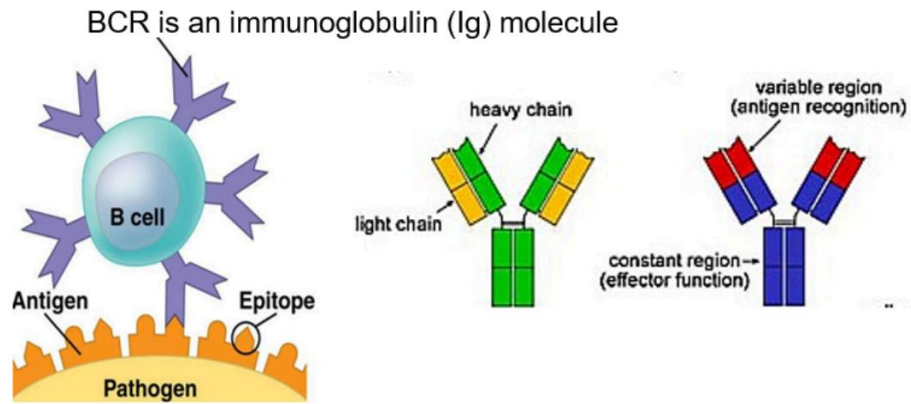
BCR is an immunoglobulin (Ig) molecule

structure: 2 identical heavy chains & 2 identical light chains; see right image for constant region and variable regions

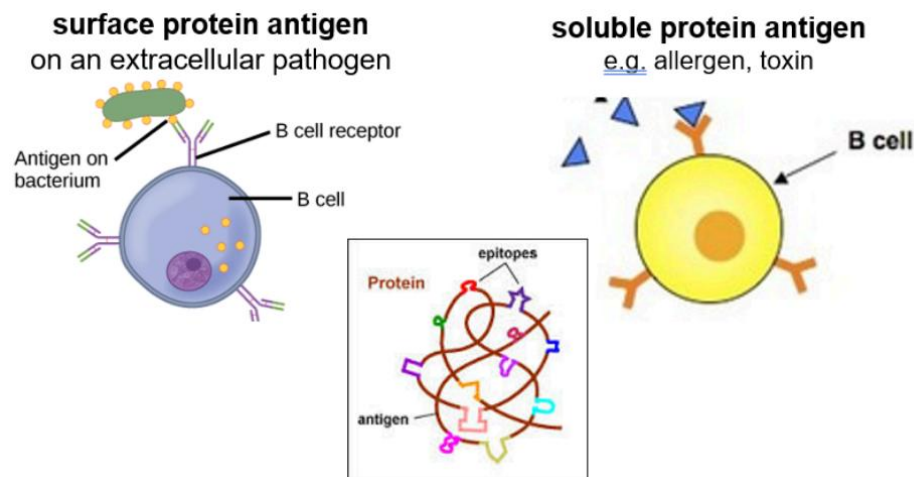
Ig molec expresses 2 identical variable regions - can bind to the same antigenic epitope twice

Can bind directly to antigenic epitope, unlike T cell receptor

TCR can only bind to one peptide (protein antigens); BCR can bind directly to both a protein and non-protein antigens



Most common B cell response is against protein antigens



Epitopes are where the variable region of B cell receptor and antibody will bind to .

Reviewing a primary immune response

With B cell immunity, the naive B cell is circulating through the secondary lymphoid tissues -> if it finds its unique protein antigenic epitope it will bind directly to it (could be on extracellular pathogen or soluble protein)

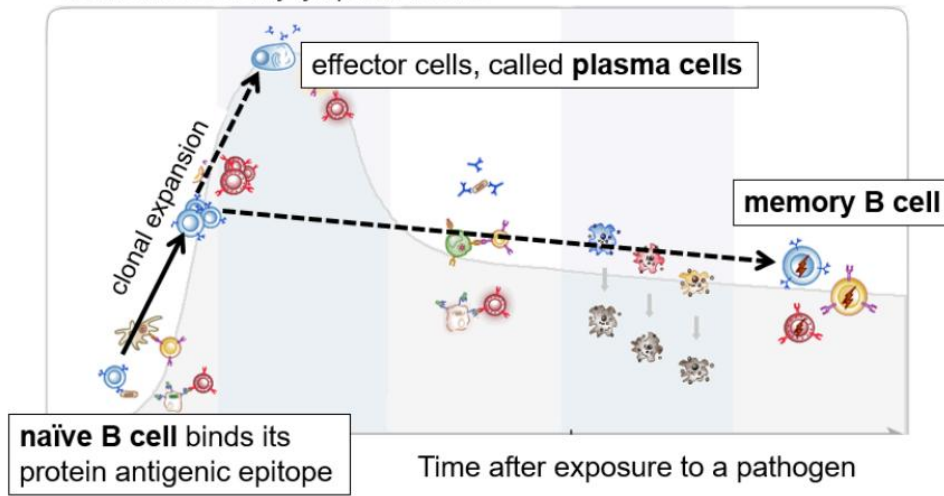
Naive B cell then undergoes clonal expansion

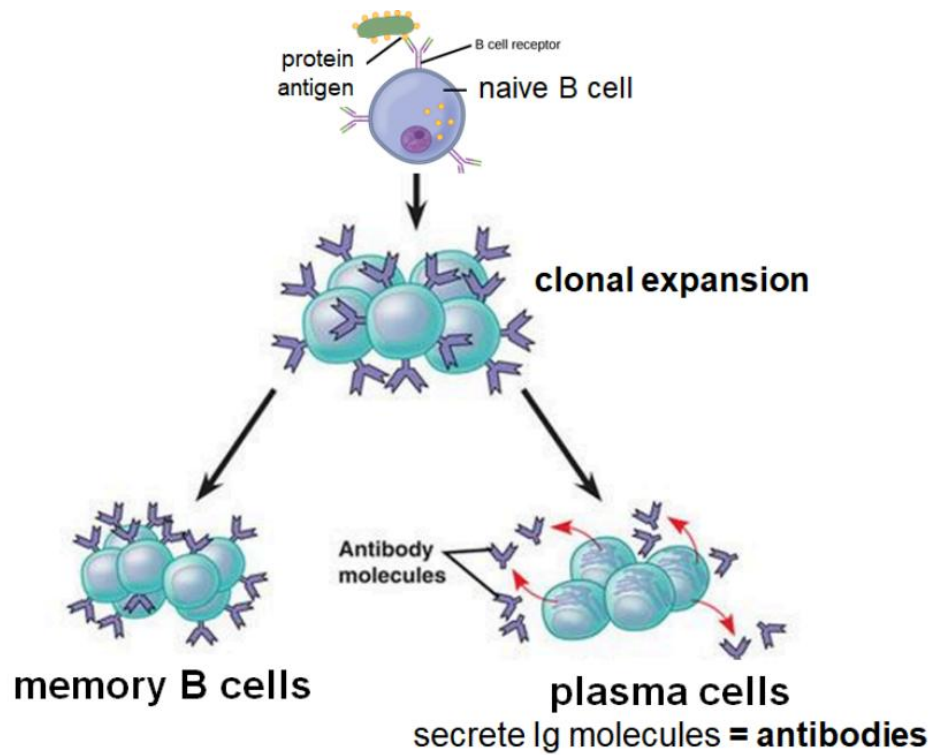
Daughter cells will become either effector cells or memory B cells

Different from T cell immunity where all daughter T cells become effector cells

Primary B cell response to a protein antigen generates plasma cells and memory B cells

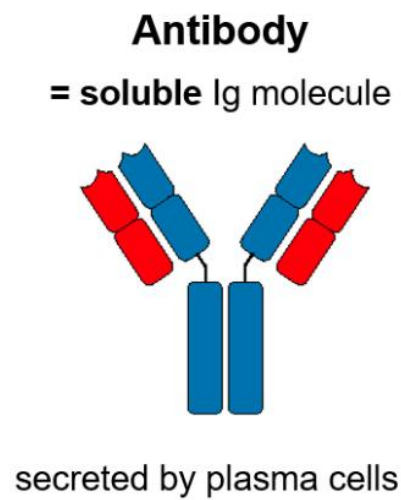
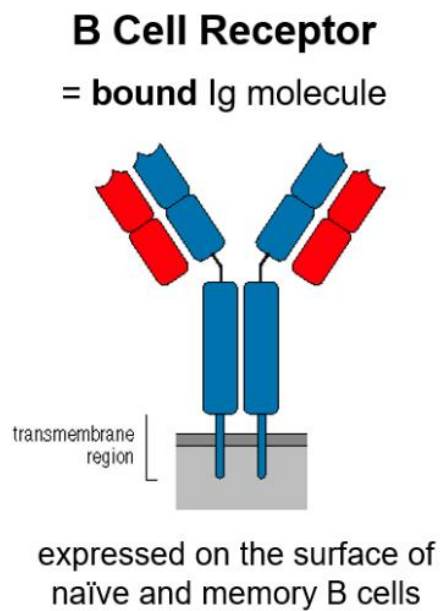
occurs in secondary lymphoid tissues





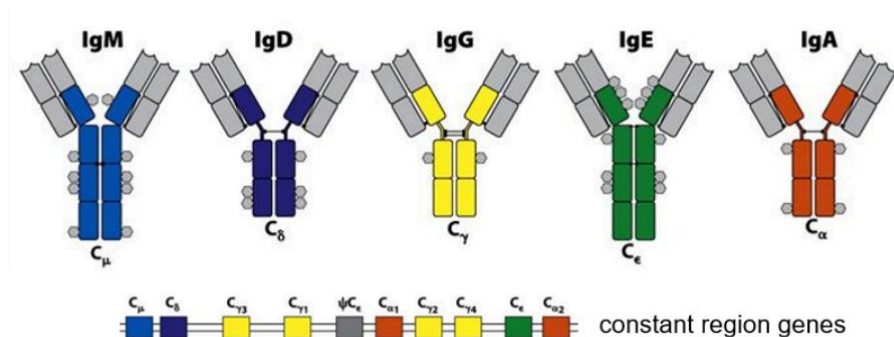
Plasma cells secrete immunoglobulin molecules called antibodies - Secretes approx 2000 antibodies per sec for 3-5 days - whether daughter cell becomes memory or plasma cell not well understood

Structure of B cell receptor and antibody



B cell receptor and antibody are both immunoglobulin molecules - the difference is that B cell receptor is bound to the surface of B cell (naïve or memory) while antibody is a soluble Ig molecule (secreted by plasma cells)

5 classes of immunoglobulin molecules



Class is defined by differences in **heavy chain constant region**

- Length
- Distribution of carbohydrates
- Location of hinges (disulphide bonds)

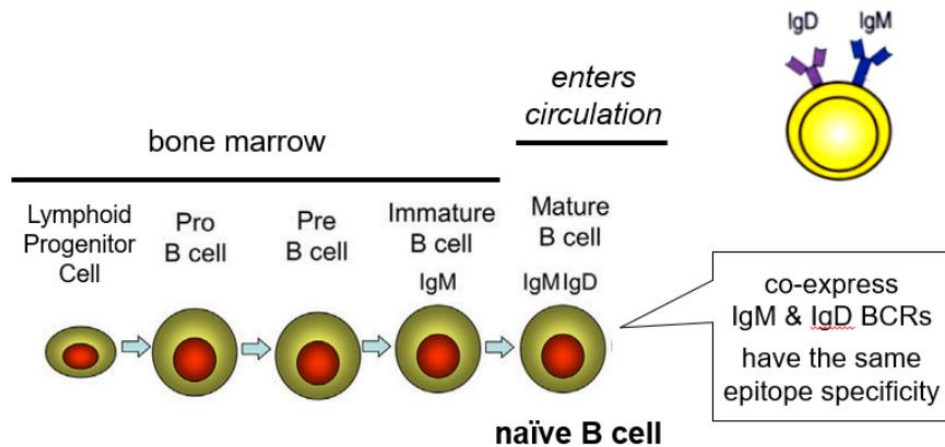


- molecules have diff gene regions w/in heavy chain constant region

- hinges gives the molecule flexibility so it can flex and bind to its antigenic epitope at a higher affinity than a ig molecule can

IgE (see image above) do have a role in immunity against bacteria and viruses - instead it is important against parasites

B Cell Development



When B cell is developing, it starts as a hematopoietic stem cell, then lymphoid progenitor cell, then slowly develops into B cell (see image)

Once B cell enters into circulation, it is a naïve B cell and co-expresses both IgM & IgB B cell receptor - IgM and IgB receptor have exactly the same variable

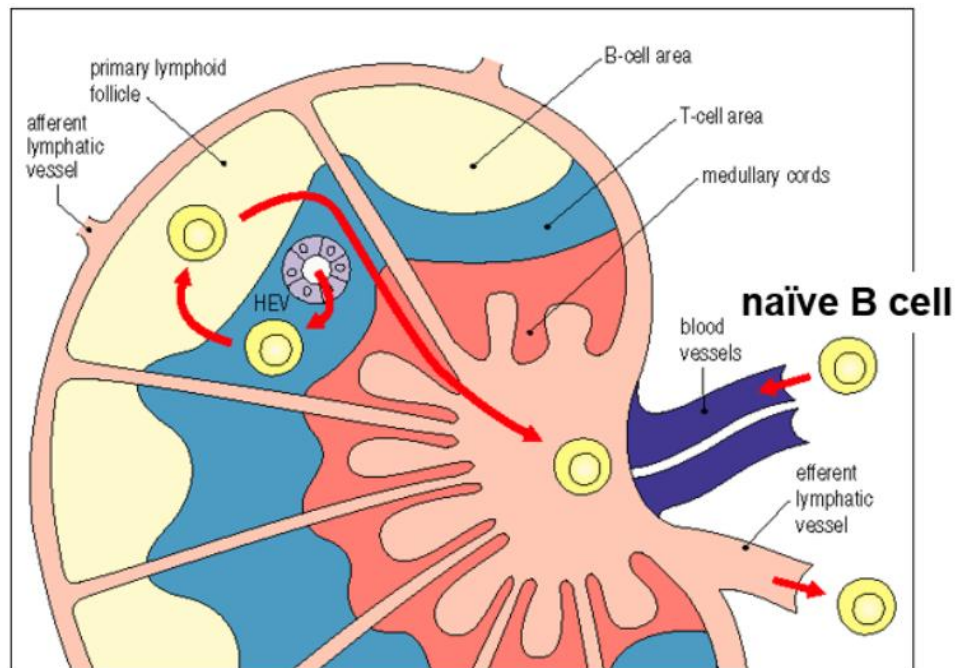
region - have exactly same specificity for same epitope (recall: one B cell has specificity for one unique antigenic epitope)

- Why B cell expresses both receptors remain unclear

Topic 8-2

Like T cells, B cells circulate thru SLT

If it binds to its antigen it remains inactive and will re-circulate for a few months. If it comes into contact with its antigen w/in that time it will undergo apoptosis.

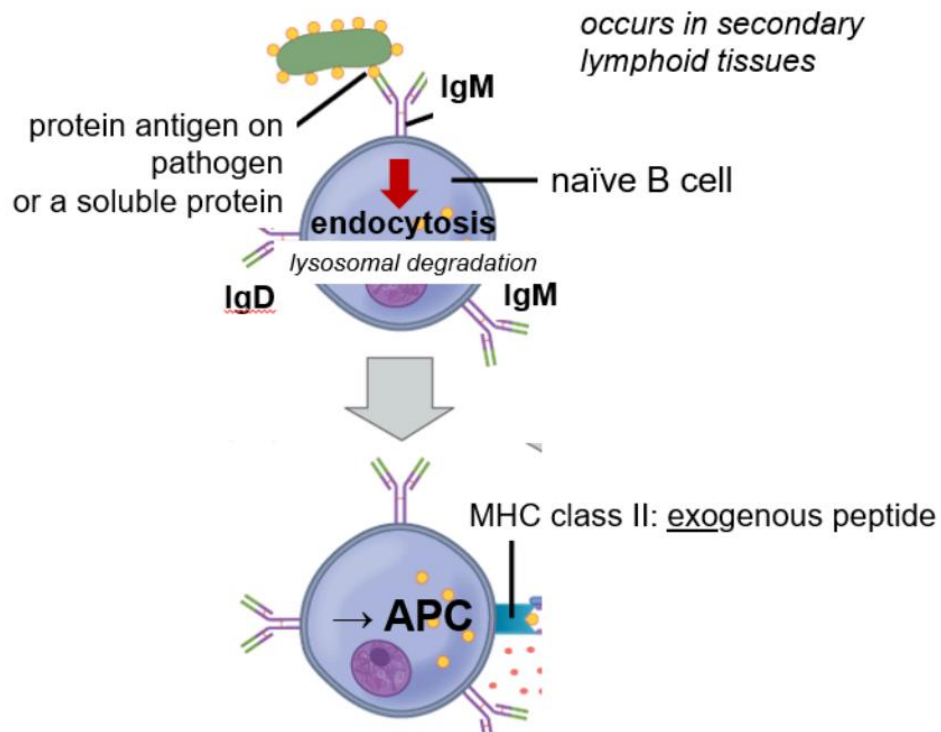


Most common B cell responses is against protein antigens

Surface protein antigen on an extracellular pathogen

Soluble protein antigen ie: allergen

B cell immune responses to a protein antigen



If naive B cell meets its specific protein antigen or extracellular pathogen or a soluble protein, that naive B cell will bind it by its variable region and then internalize it via endocytosis - Recall: endocytosis -> lysosomal degradation so exogenous peptides generated will be presented by that B cell on MHC class 2 molecules

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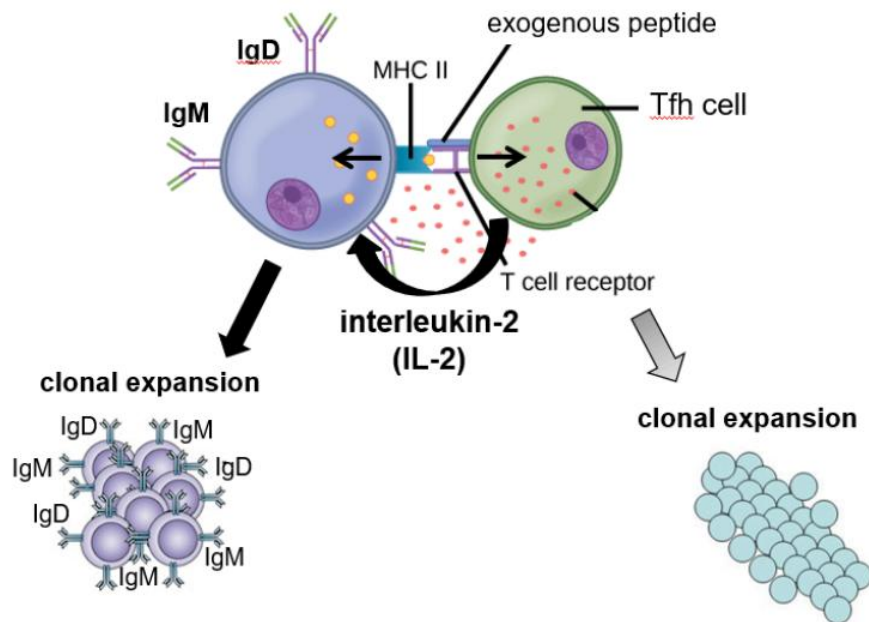
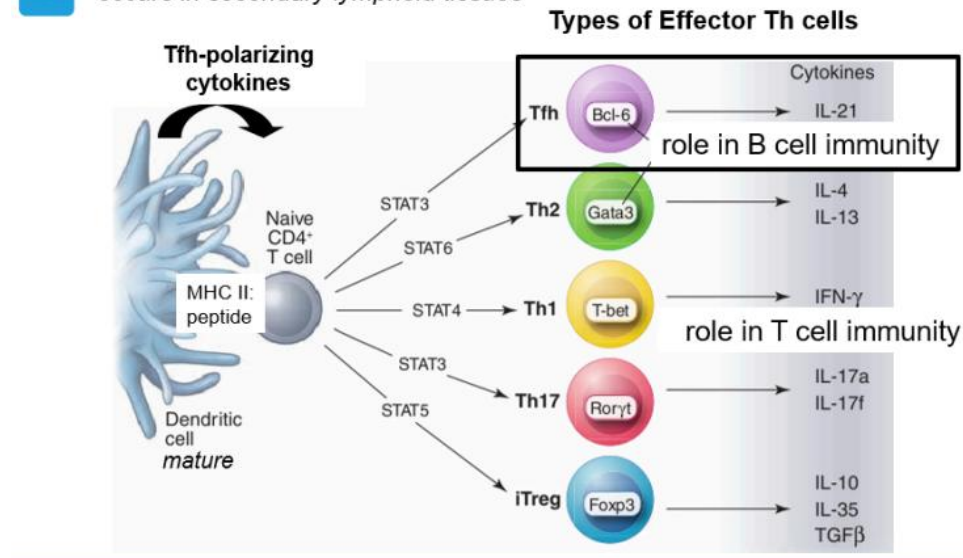
- Recall: B cells are key antigen presenting cells

B cell immunity against a protein antigen requires help from T follicular helper

Cells

RECALL

occurs in secondary lymphoid tissues



Tfh cytokines activate clonal expansion

Tfh has a T cell receptor w/ a variable region specific for that peptide being presented by the B cell

Binding interaction activates both the B cell and Tfh cell

Tfh cell begins to secrete interleukin-2 (IL-2)

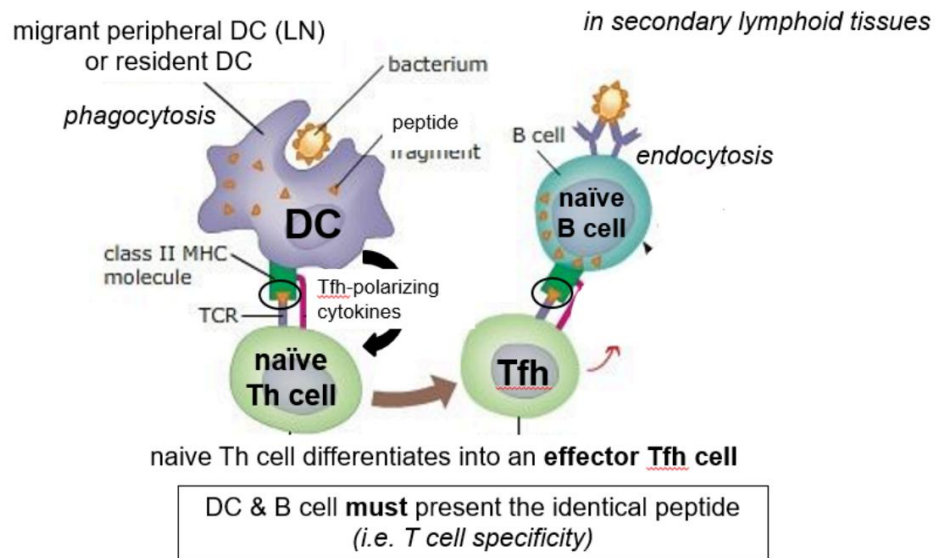
IL-2 activates B cell clonal expansion

Tfh will also undergo clonal expansion

T cell immunity must occur to initiate B cell immunity

This all happens in SLT

Summary



- Both DC

and naïve B

cell

Consumption

The same is the same as the same .

pathogen

order to

degrades

They and

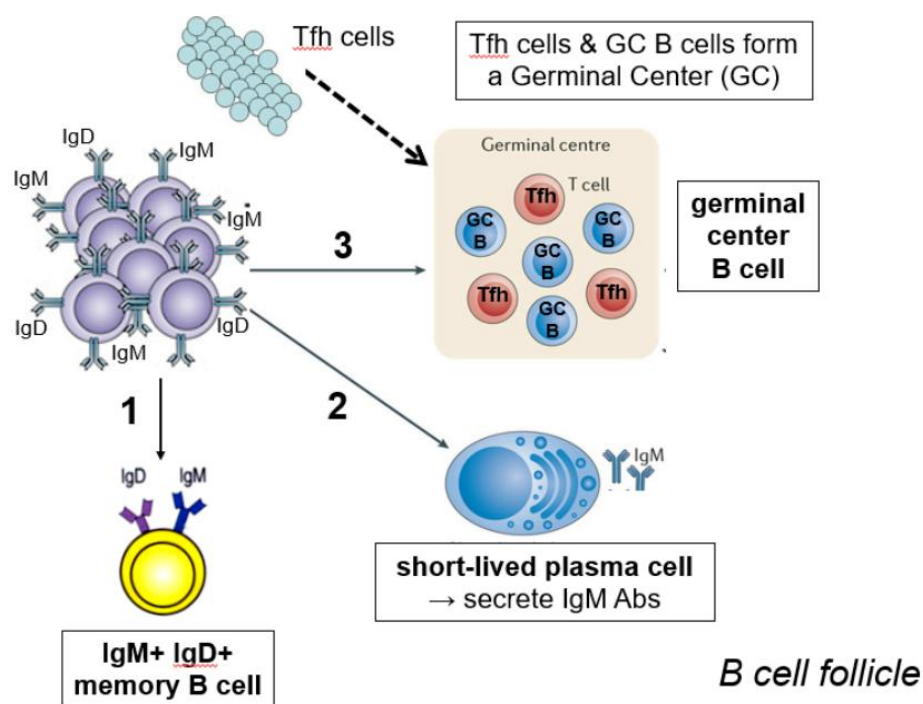
Use them

peptides

Recognition

T cell immunity must be initiated first in order to activate B cell immunity (bc Tfh cells)

The 3 fates of daughter B cells



Like naive, daughter cells also expressing IgM and IgB

Some remains as IgM & IgB

Memory B cells - Some become short lived plasma cells these cells secrete IgM antibodies - Daughter cell and Tfh cells form a germinal center (GC)-> Germinal center B cells - GC only forms during a immune response , when its stopped GC will disappear

B cells

Center B cells

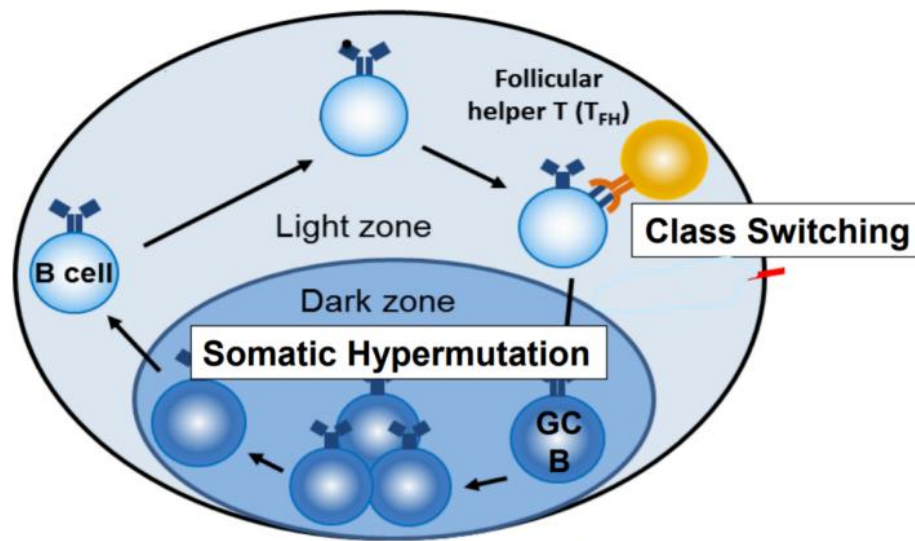
For the next vids, watch them while taking no/minimal notes.

Topic 8-3

2 GC events that occur are Class switching and Somatic Hypermutation

SM alters the regions genes

CS alters heavy chain and constant regions



alters **variable region** genes

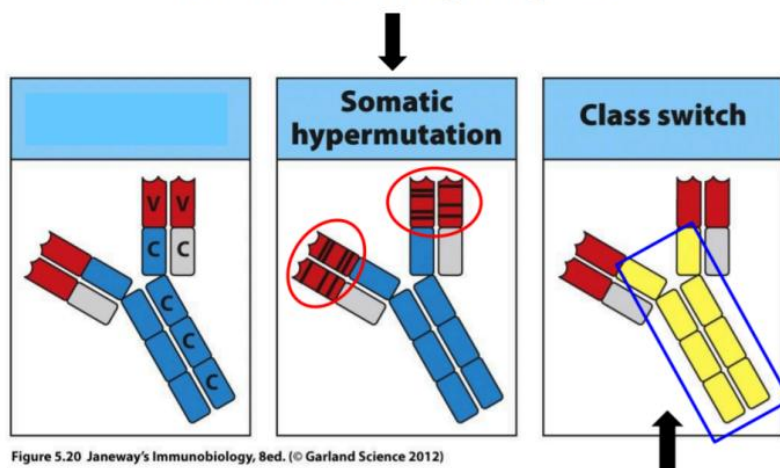


Figure 5.20 Janeway's Immunobiology, 8ed. (© Garland Science 2012)

alters **heavy chain
constant region** genes

SHM incre variable regions affinity for its epitope

The variable region remains the same

